Kejing Xia

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Education

Georgia Institute of Technology *Aug. 2025 – Present*
Ph.D. Student in Computer Science
- Advisor: Prof. Wenke Lee

Wuhan University *Sep. 2021 – June 2025*
B.Eng. in Communication Engineering (Artificial Intelligence Track)
- GPA: 3.90/4.00, Rank: Top 5%, First-Class Academic Scholarship

Research Interests

My research focuses on **understanding and improving reasoning in language models**. I study useful structure in model computation, reasoning trajectories, and interaction data, and use these insights to guide model design, post-training, and inference algorithms.

- **Reasoning and Inference Mechanisms**: MetaState and RoT study how information flow and process signals can guide reasoning. LongMamba and R²-dLLM use model dynamics to improve long-context inference and diffusion decoding.
- **Learning from Interaction and Task Structure**: LookAgain-CUA learns when agents need new observations, while SourceLearn uses task feedback to refine reusable source understanding. GEvol uses task dependencies to evolve executable benchmark tasks.

Experience

Georgia Institute of Technology, GTISC Lab | Ph.D. Student *Jan. 2026 – Present*
Advisor: Prof. Wenke Lee
Research Focus: *Reasoning, Agents, and LLM Post-Training*
- **MetaState** [**COLM'26**] equips a frozen dLLM with lightweight recurrent working memory, allowing continuous information to persist across denoising steps rather than passing only discrete tokens.
- **Rhythm of Thought** [**Preprint**] is an RL-trained latent reasoning framework that predicts a Gaussian over trace-relative progress. The resulting Gaussian phase signal modulates decoder layers at each reasoning step.
- **SourceLearn** [**Preprint**] turns repeated access to a persistent source into reusable source understanding through self-directed study and task-guided refinement of an external source model.

Zoom Communications, Inc., AI Incubation | Research Intern *May 2026 – Aug. 2026*
Mentor: Dr. Renqian Luo
Research Focus: *Efficient Computer-Use Agents*
- **LookAgain-CUA** [**Preprint**] learns when multiple GUI actions can be decided from a single observation. BoundaryTrace provides multi-action supervision for self-refined SFT, and cost-aware RL refines the policy.

Georgia Institute of Technology, EIC Lab | Undergraduate to Ph.D. Student *May 2024 – Dec. 2025*
Advisor & Mentors: Prof. Yingyan (Celine) Lin , Yonggan Fu , Zhifan Ye
Research Focus: *Efficient Diffusion Decoding and Long-Context Modeling*
- **LongMamba** [**ICLR'25**] identifies local and global channels in Mamba, traces long-context failure to state decay in global channels, and enlarges their receptive fields through training-free token filtering.
- Related work includes R²-dLLM [**Preprint**] for redundancy-aware diffusion decoding, training-free soft token interpolation for masked dLLMs, and LAMB [**ACL'25**] for attention-guided token filtering in SSMs.

Bosch Research & Innovation Software Development | Research Intern *July 2024 – May 2025*
Mentor: Dr. Youjian Zhang
Research Focus: *Geometric Prior Enhanced Urban Scene Reconstruction*
- **D²GS** [**NeurIPS'25**] reconstructs urban scenes without LiDAR by iteratively co-optimizing diffusion-generated dense depth and 3D Gaussian Splatting, with progressive pruning and road-specific geometric constraints.

Wuhan University, Signal Processing Lab | Undergraduate Researcher *Oct. 2023 – Apr. 2024*
Advisor: Prof. Lei Yu
Research Focus: *3D Vision and Event-based Vision*
- **BRS-E²NeRF** [**PR'26**] combines events and RGB images through event-enhanced re-blurring and double-cycle optimization to jointly correct motion blur and rolling-shutter distortion for 3D reconstruction.

Publications

** equal contribution*

[1] **K. Xia**, Z. Du, S. Yao, L. Fu, H. Shen, X. Ma, L. Wei, M. Li, X. Yuan, D. Shi, Q. Jin, B. A. Prakash, W. Lee, and R. Luo. When to Look Again: Learning Observation Boundaries for Efficient Computer-Use Agents. Submitted to ACL ARR 2026.
[2] Z. Du, **K. Xia**, J. Zhang, S. Yao, Z. Shi, Z. Zheng, J. Luo, X. Yuan, D. Shi, Q. Jin, and W. Lee. GEvol: Graph-Guided Benchmark Evolution for Terminal Agents. Submitted to ACL ARR 2026.
[3] **K. Xia**, L. Wei, M. Li, D. Shi, X. Yuan, Z. Du, Q. Jin, and W. Lee. Rhythm of Thought: Progress-Aware Latent Reasoning via Reinforcement Learning. Submitted to ACL ARR 2026.
[4] **K. Xia**, M. Li, L. Wei, Z. Du, X. Yuan, D. Shi, Q. Jin, and W. Lee. MetaState: Persistent Working Memory Enhances Reasoning in Discrete Diffusion Language Models. COLM 2026. [Paper] [Code] [Website]
[5] Z. Du, **K. Xia**, X. Zhong, Y. Fu, N. Oswald, B. Ji, B. Khailany, P. Molchanov, and Y. Lin. R²-dLLM: Accelerating Diffusion Large Language Models via Spatio-Temporal Redundancy Reduction. Preprint, 2026. [Paper] [Code]
[6] L. Fu, **K. Xia**, Y. Wang, Y. Jin, J. He, X. Yang, H. Liu, Y. Yu, H. Jin, Y. Xiao, W. Lee, B. A. Prakash, and H. Wang. From Knowledge Access to Source Learning: Developing Source-Specific Competence. Preprint, 2026. [Paper] [Code] [Website]
[7] Q. Jin, L. Tung, K. Li, Q. Shi, Y. She, H. Jia, H. Zhao, **K. Xia**, Z. Du, et al. ChainSWE: Benchmarking Coding Agents on Multi-Bug Software Maintenance. EMNLP Main 2026. [Paper] [Benchmark]
[8] Z. Du*, J. Luo*, Z. Zheng, X. Yuan, **K. Xia**, D. Shi, Q. Jin, Q. He, S. Zou, Y. Liang, and W. Lee. PACT: Privileged Trace Co-Training for Multi-Turn Tool-Use Agents. Preprint, 2026. [Paper] [Website]
[9] D. Shi, H. Zhu, X. Yuan, W. Zhao, **K. Xia**, W. Xiao, and W. Lee. CopT: Contrastive On-Policy Thinking with Continuous Spaces for General and Agentic Reasoning. Preprint, 2026. [Paper]
[10] X. Yuan, X. Diao, Z. Liu, C. Zhang, D. Shi, **K. Xia**, Z. Du, S. Yao, S. Vosoughi, and W. Lee. Priority-Preserving Advantage Aggregation for Multi-Reward Policy Optimization. Preprint, 2026.
[11] Y. Peng, M. Li, **K. Xia**, R. Zhang, and A. Houmansadr. Membership Inference Attacks on Vision-Language-Action Models. Preprint, 2026. [Paper]
[12] **K. Xia***, J. Jia*, K. Jin, Y. Bai, L. Sun, D. Tao, and Y. Zhang. D²GS: Dense Depth Regularization for LiDAR-free Urban Scene Reconstruction. NeurIPS 2025. [Paper] [Code] [Website]
[13] Z. Ye*, **K. Xia***, Y. Fu, X. Dong, J. Hong, X. Yuan, S. Diao, J. Kautz, P. Molchanov, and Y. Lin. LongMamba: Enhancing Mamba's Long-Context Capabilities via Training-Free Receptive Field Enlargement. ICLR 2025. [Paper] [Code] [Website]
[14] M. Li, **K. Xia**, G. Zhang, Z. Wang, G. Tao, S. Pan, J. Zhai, and S. Ma. EDITOR: Effective and Interpretable Prompt Inversion for Text-to-Image Diffusion Models. Preprint, 2025. [Paper]
[15] Z. Ye, Z. Wang, **K. Xia**, J. Hong, L. Li, L. Whalen, C. Wan, Y. Fu, Y. Lin, and S. Kundu. LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering. ACL Main 2025. [Paper] [Code]
[16] H. Wang*, **K. Xia***, Y. Wang, C. Zhang, M. Lin, and L. Yu. BRS-E²NeRF: Event Enhanced Neural Radiance Fields from Blurry Rolling Shutter Images. Pattern Recognition, 2026. [Paper]
[17] **K. Xia***, L. Wei*, and L. Yu. A Spatio-temporal Event Transformer on Versatile Tasks for Human Behavior Analysis. MiGA Workshop at IJCAI 2024. [Paper]

Honors and Awards

- **Top Reviewer**, NeurIPS 2026 Main Conference — *2026*
- **National Second Prize (Team Captain, Fifth Place)**, RoboCup China Open — *2024*
- **First-Class Academic Scholarship**, Wuhan University — *2023 – 2024*
- **"Luojia" Distinguished Exchange Student Scholarship**, Wuhan University — *2023 – 2024*
- **National-Level Project**, Student Innovation & Entrepreneurship Project — *2023 – 2024*
- **National First Prize**, Huazhong Cup Mathematical Modeling Challenge — *2023*
- **Hubei Province First Prize**, China Undergraduate Mathematical Contest in Modeling — *2023*
- **Outstanding Student**, Wuhan University — *2022 – 2023*
- **Merit Student**, Wuhan University — *2021 – 2022, 2023 – 2024*

Teaching and Service

- **Teaching Assistant**, CS 6035 Introduction to Information Security, Georgia Tech — *Spring 2026*
- **Teaching Assistant**, CS 6262 Network Security, Georgia Tech — *Spring 2026*
- **Teaching Assistant**, CS 4803/8803 Efficient Machine Learning, Georgia Tech — *Fall 2025*
- **Conference Reviewer**, ECCV, COLM, NeurIPS, EMNLP, ICLR, NAACL — *From 2026*